CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 46 (2014) 103–107

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Experimental and thermodynamic assessment of the Nd–Zr system

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ARTICLE INFO

Article history:
Received 29 January 2014
Received in revised form
25 February 2014
Accepted 28 February 2014
Available online 13 March 2014

Keywords:
Nd–Zr phase diagram
CALPHAD
Thermodynamic assessment
In situ X-ray diffraction
Conical nozzle levitation

ABSTRACT

The phase equilibria of the binary Nd–Zr system have been investigated by combining the gas levitation technique with *in situ* high energy synchrotron X-ray diffraction at elevated temperatures. The Nd–Zr system is of a eutectic type. From the diffraction data the liquidus line was experimentally determined between 1300 and 1800 K. At $T=1274\pm10$ K the eutectic reaction L↔ *bcc*-Nd+*bcc*-Zr is evident. The eutectic composition $x=7\pm2$ at% Zr is determined from the analysis of the microstructure of cast alloys in accordance with the *in situ* diffraction data. Based on the experimental data the Nd–Zr phase diagram is presented and a self-consistent thermodynamic description is given.

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1. Introduction

Rare earth-transition metals (RE-TM) are used as permanent magnets which derive their exceptional magnetic properties from the favorable combination of rare earth metals with transition metals in particular alloys based on $(Nd,Pr)_2Fe_{14}B$ and $Sm_2(Co,Cu,Fe,Zr)_{17}$ [1]. For further improvement of the properties by designing the microstructure, the knowledge of thermodynamic data is indispensable. Calculation of the corresponding phase diagrams by the CALPHAD method [2] is a powerful tool for the optimization of alloys and thermal processing. The thermodynamic description of such multi-component systems requires all subsystems. However, for the binary Nd–Zr system there is no phase diagram reported in [3,4]. One difficulty is the high chemical reactivity of RE-TM melts with any container, which gives rise to errors in the experimental data, especially for the phase equilibrium with the liquid phase. This problem can be overcome by combining the container-less levitation technique with *in situ* high energy synchrotron X-ray diffraction at elevated temperatures. An overview of different types of levitation techniques and its use for studying structure and thermo-physical properties of liquids can be found in [5]. Recently, this technique was successfully employed for the assessment of the phase diagrams Gd–Zr and Gd–Ti [5–7]. Here, we report experimental results of the binary Nd–Zr phase diagram and its thermodynamic description.

2. Experimental

For the microstructure and thermal analyses, master alloys of Nd_xZr_{100-x} ($x=5$, 10, 15, and 60) with a mass of about 20 g were prepared from pure elements 99.98 wt% Zr (HWN titan GmbH), and 99.9 wt% Nd (Goodfellow) in an arc furnace, on a water-cooled copper hearth under purified argon atmosphere. The buttons were remelted in a Hukin-type cold-crucible induction melting facility and cast into a copper mold with 6 mm inner diameter. Cross-sections of cast rods were prepared for metallographic examination by polishing resin-mounted specimens with diamond pastes. The microstructure was analyzed by scanning electron microscopy (SEM) in the backscattered electron (BSE) mode using a Gemini-Zeiss microscope equipped with an energy-dispersive spectrometer (Bruker AXS) for composition analysis. The chemical composition of the phases was determined at 30 keV using the intensities of the Nd L- and Zr K-emission lines. Thermal analysis was carried out by differential scanning calorimetry (DSC, Netzsch) at a heating rate of 20 K/min. For *in situ* X-ray diffraction

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http://dx.doi.org/10.1016/j.calphad.2014.02.008
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investigations, alloy ingots of about 2 mm in diameter (40 mg mass) were prepared with nominal composition $Nd_{40}Zr_{60}$ by arc-melting proper quantities of pure Nd and Zr under an argon atmosphere with a Ti getter. The ingots were inverted and remelted several times to ensure compositional homogeneity. The difference in sample mass after arc-melting was less than 0.1%. For calibration also pure elements Nd and Zr samples were prepared as 2 mm sphere-like samples.

In situ high energy synchrotron X-ray diffraction (XRD) was performed in transmission geometry (wave length $\lambda = 0.01093$ nm) at the beam-line BL04B2 of SPring-8 in Japan. The samples were processed by conical nozzle levitation using filter-purified Ar gas (impurity < 0.01 mass ppm). A laser has been applied for heating the samples. During the experiments, the levitated sample was repeatedly heated above its liquidus temperature ($T_{liquidus}$) and subsequently cooled down passing solidification and solid-state transformations to ensure reproducibility of the measurements. The temperature was measured by a two-color pyrometer. The pyrometer values for temperatures where liquid phase was present were corrected by the calibration of the melting visible in the X-ray diffraction patterns ($T_{XRD} = 1333$ K) to the corresponding DSC event ($T_{melt} = 1274$ K) according to $T_{corr} = 1/(\frac{1}{T_{pyro}} + \frac{1}{T_{melt}} - \frac{1}{T_{XRD}})$. The temperatures of the solid state transformation seen in XRD data were in agreement with the DSC measurement within ± 20 K. The scattered intensity was continuously recorded with acquisition time of 0.6 s by a two-dimensional X-Ray Image Intensifier with a coupled CCD-camera during heating or cooling. More details on the set-up can be found in [8]. The scattering intensity $I(q)$ as a function of the wave vector q ($q = 4\pi \sin\theta/\lambda$) was obtained by Azimuthal integration of the original 2D images using Fit2D software package [9].

3. Experimental results

Fig. 1 shows the DSC scan of the $Nd_{40}Zr_{60}$ alloy upon heating up to 1773 K. The curve indicates three phase transformations upon heating. The on-set temperatures are given in Fig. 1. The corresponding phase reactions can be determined unambiguously by the *in situ* XRD patterns. At room temperature Nd has a double hexagonal close-packed structure (*dhcp*-Nd) which is different in point symmetry from the hexagonal room temperature phase of Zr (*hcp*-Zr). At high temperatures both elements transform into the cubic body-centred crystal structure. Fig. 2 shows selected

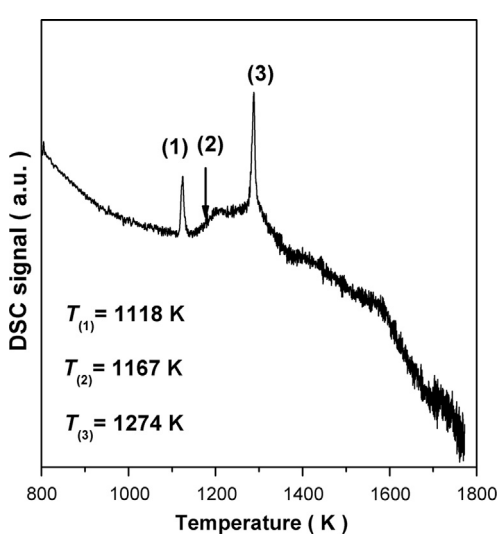

Fig. 1. DSC scans of $Nd_{40}Zr_{60}$ upon heating up to 1773 K (heating rate 20 K/min).

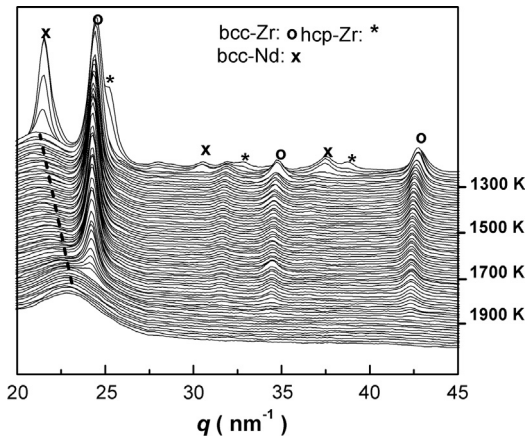

Fig. 2. XRD patterns of $Nd_{40}Zr_{60}$ at elevated temperatures (the given temperatures are only approximated values due to non-equidistant temperature steps). The positions of reflections of the crystalline phases are indicated by the different symbols. The dashed line marks roughly the position of the first diffuse maxima of the Nd(Zr) liquid phase.

diagrams $I(q)$ of the $Nd_{40}Zr_{60}$ alloy at different temperatures representing the characteristic phase equilibria. The whole set of recorded XRD patterns enables to determine the different coexistence ranges of phases as a function of temperature. Below $T = 1118$ K, *dhcp*-Nd(Zr) and *hcp*-Zr(Nd) coexist. The reaction (1) of the DSC curve at $T = 1118$ K corresponds to the transformation *dhcp*-Nd(Zr) + *hcp*-Zr(Nd) ↔ *bcc*-Nd(Zr). Up to $T = 1130$ K we observe the reflections of *bcc*-Nd(Zr) and *hcp*-Zr(Nd). At $T = 1167$ K, DSC event (2), the transformation *hcp* Zr(Nd) ↔ *bcc*-Zr(Nd) + *bcc*-Nd(Zr) takes place. The DSC signal is rather broad probably due to difference in solubility of Nd in both allotropes and by the kinetics determined by diffusion of Nd in Zr. Between 1200 K < T < 1270 K we observe the equilibrium between *bcc*-Nd(Zr) and *bcc*-Zr(Nd) phases. The DSC event (3) at $T = 1274$ K corresponds to the eutectic reaction L ↔ *bcc*-Nd(Zr) + *bcc*-Zr(Nd). The melting of the sample can be seen by the disappearance of the *bcc*-Nd(Zr) reflections and the occurrence of a diffuse diffraction peak in the XRD-patterns. Above $T > 1274$ K the Nd-rich melt is in equilibrium with the *bcc*-Zr(Nd) phase. For $Nd_{40}Zr_{60}$ upon heating up to $T = 1850$ K, the fraction of *bcc*-Zr decreases with temperature, and above $T_{liquidus} = 1850$ K $\pm$ 20 K, only liquid phase is present (Fig. 2). In the semi-solid regime one can observe a systematic shift of the position of the diffuse maxima (dashed line in Fig. 2).

The positions of the diffuse maxima of the liquid phase and that of the (110)-reflection of the *bcc*-Zr phase were calculated by a fit of two Lorentzian curves. Fig. 3 shows the temperature dependence of both q-positions together with data obtained for the elements (pure Nd liquid and (110) reflection of pure Zr). The presented data were obtained from two independent measurements of different heating cycles in order to prove the reproducibility. The shift of q_1 of the Nd(Zr) melt towards larger q-values reflects mainly the change in composition with temperature and contains the information about the liquidus line. For metallic glasses and metallic liquids the position of the first diffuse maximum q_1 scales approximately with a concentration-averaged atomic diameter d of the alloy analogously to the Ehrenfest-equation $q_1 = 1.22 \times 2\pi/d$ [10] with $d = \sum c_i d_i$, where d_i is the atomic diameter of element i and c_i is the concentration of element i. This equation is not exactly valid, however, within an element substitution series the change in q_1 can be often well described by a linear dependence on the concentration c_i of the added element [11,12]. The experimentally observed shift of the position of the maxima to larger q-values with rising temperature can be understood by an increased solubility of Zr ($d_{Zr} = 0.316$ nm) in the Nd-rich liquid ($d_{Nd} = 0.366$ nm)

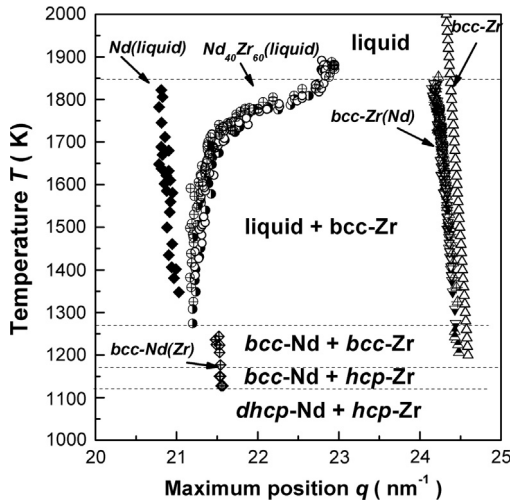

Fig. 3. Temperature dependence of diffraction maxima of $Nd_{40}Zr_{60}$. (a) Nd(Zr) liquid (circles), (b) (110)-reflection of *bcc*-Zr(Nd) (triangles), and (c) (110)-reflection of *bcc*-Nd(Zr). Also shown are the positions of the pure elements. (d) Nd-liquid and (e) *bcc*-Zr. The differences between (a) and (d) as well as (b) and (e) are due to the solubility of Zr in the Nd-rich liquid, and Nd in the *bcc*-Zr lattice, respectively.

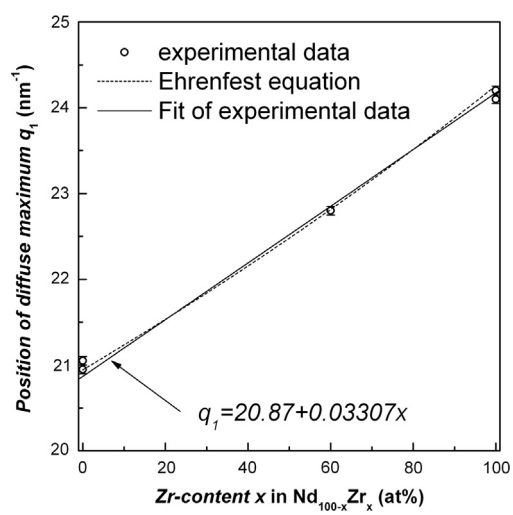

Fig. 4. Composition dependence of the position of the first diffuse maximum q_1 of $Nd_{100-x}Zr_x$ liquids (x=0, 40, and 100). Analytical functions by a linear fit (line) and by Ehrenfest-equation are given.

which reduces the average atomic diameter. Fig. 4 shows experimentally determined values for the melts of Nd, Zr, and $Nd_{40}Zr_{60}$ as well as the calculation using the Ehrenfest-equation. From these data we can conclude that the concentration dependence of the maximum position q_1 versus Zr content x can be well approximated by a linear fit of the experimental data as given in Fig.4. The thermal expansion of the liquids has also an impact on the maxima positions as can be seen by the behavior of $q_{Nd\text{-}liquid}$ of the pure Nd melt in Fig. 3. We observe a small shift towards lower q-values which can be approximated by $q_{Nd\text{-}liquid}=(21.542-4.12\times10^{-4}T)$ nm^{-1}. Assuming the same thermal coefficient for the Nd–Zr liquid, we can correct the effect of volume expansion on the maximum position in order to extract the compositional contribution. Using the composition dependence according to the fit function shown in Fig. 4, we can calculate the Zr-content x of the Nd(Zr) liquid at a given temperature from the experimental data $q_1(T)$ in nm^{-1} corrected for the contribution of thermal expansion

$$x_{Zr}^{liquid}(T)=\frac{q_1(T)-21.542+4.12\times10^{-4}T}{0.03307} \quad (1)$$

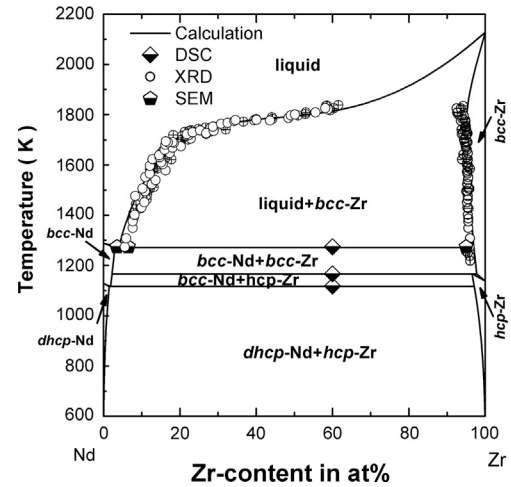

Fig. 5. Nd–Zr phase diagram. Experimental data from *in situ* XRD, DSC and SEM by marks. Calculation using thermodynamic description of this work is represented by the lines.

By applying Eq. (1) for the values of the liquid diffraction maximum in the two phase region L+*bcc*-Zr(Nd), the calculated compositions then represent the liquidus line of the Nd–Zr phase diagram, as shown in Fig. 5. From the temperature dependence of the liquidus line $x_{Zr}(T)$, the eutectic composition can be determined as 5.5 ± 2 at% Zr.

Similar to the liquid phase one can also calculate the solubility of Nd in the *bcc*-Zr(Nd) phase by the analysis of the difference of the peak positions to those of the pure *bcc*-Zr. The shift of the *bcc*-Zr (110)-reflection $q_{(110)}$ with temperature towards lower q-values is related to the thermal expansion but is additionally biased by the temperature dependent solubility of Nd. Using Vegard's rule for a solid solution between *bcc*-Nd and *bcc*-Zr, the concentration of Nd (in at%) of the *bcc*-Zr(Nd) phase is given by

$$x_{Nd}(T)=\frac{q_{bcc-Zr(Nd)}(T)-q_{bcc-Zr}(T)}{q_{bcc-Nd}-q_{bcc-Zr}}\times100\% \quad (2)$$

The temperature dependence of the q-value of the pure *bcc*-Zr phase was calculated by an extrapolation of the literature data lattice parameter given in [13,14], also presented in Fig. 4. For the difference $\Delta q=q_{bcc-Nd}-q_{bcc-Zr}$ of the pure *bcc*-phases a constant value $\Delta q=3.03$ nm^{-1} is used according to values $a_{bcc\text{-}Zr}=0.3609$ nm at $T=1135$ K [13] and $a_{bcc\text{-}Nd}=0.413$ nm at $T=1156$ K [14], assuming that the influence of slightly different thermal expansion coefficients can be neglected. A maximum solubility of 5 ± 1 at% Nd in *bcc*-Zr(Nd) at $T=1700$ K is obtained from the diffraction data. The calculated lattice parameter of the *bcc*-Nd (Zr) phase at $T=1150$ K is within the error limits identical to the value for pure Nd ($a=0.413$ nm at $T=1156$ K [14]), which means a very low solubility of Zr in *bcc*-Nd. Fig. 5 also comprises the results of phase composition and transformation temperatures as determined by the DSC measurements, *in situ* XRD, and those of the microstructure analysis discussed in the following.

Fig. 6 shows the SEM image of the cast $Nd_{90}Zr_{10}$ alloy. The microstructure clearly reveals the primary crystallization of Zr dendrites (dark) as well as eutectic regions. In order to obtain an approximate value of the eutectic composition, EDX analysis was done on sections of the eutectic cells by averaging over an area of 2 μm × 2 μm. The average composition was 7.6 ± 1.0 at% Zr which is in agreement with the XRD results. About 4.0 ± 1.0 at% Nd is found to be dissolved in the Zr dendrites. On the other side about 4.5 ± 1.0 at% Zr is dissolved in Nd grains (gray in Fig. 6). The experimentally values by SEM/EDX averaged over at least three

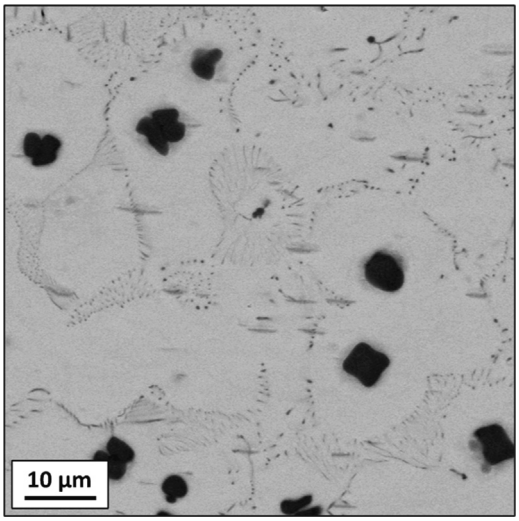

Fig. 6. Microstructure of cast $Nd_{90}Zr_{10}$ alloy around the eutectic composition. Zr-dendrites (dark) and Nd–Zr eutectic are visible. Additionally Nd grains (gray) occur due to effect of undercooling.

Table 1
Chemical compositions (at%) of primary dendrites and the eutectic of Nd–Zr as determined by EDX-analysis.

Nominal composition	Zr-dendrites	Nd-dendrite	Eutectic
$Nd_{95}Zr_5$	–	$Nd_{96.0}Zr_{4.0}$	$Nd_{92.4}Zr_{7.6}$
$Nd_{90}Zr_{10}$	$Nd_{4.4}Zr_{95.6}$	$Nd_{95.2}Zr_{4.8}$	$Nd_{92.3}Zr_{7.7}$
$Nd_{85}Zr_{15}$	$Nd_{5.0}Zr_{95.0}$	$Nd_{94.5}Zr_{5.5}$	$Nd_{92.0}Zr_{8.0}$

different sample areas are summarized in Table 1 and incorporated into Fig. 5.

4. Thermodynamic calculations

The thermodynamic assessment of the binary Nd–Zr system was done by the CALPHAD method [2]. The Gibbs free energy of each element is defined with respect to its stable magnetically disordered state at 298.15 K and 10^5 Pa (stable element reference). In the SGTE-unary database 4.4 [15], the Gibbs free energy is represented as a power series in terms of the temperature in the form

$$^0G(T) = AT\ln(T) + \sum_{n=-1} C_n T^n, \tag{3}$$

where A, C_n are coefficients and n represents a set of integers. The reference states for pure neodymium and zirconium are *dhcp*-Nd, and *hcp*-Zr, respectively. It should be noted that the data for *dhcp*-Zr are absent in SGTE pure database and difference between *hcp* and *dhcp* was arbitrary accepted as 1000 J/mol which is approximately the same value as in the Nd system.

The liquid and the disordered solid solutions of the *bcc*, *hcp* and *dhcp* phases are described using a substitutional solution model, the Gibbs free energy being defined as

$$G^{\phi} = \sum_{i=1}^{2} x_i G_i^{ref} + G^{id} + G^{ex}; \quad G^{id} = RT \sum_{i=1}^{2} x_i \ln(x_i)$$
$$G^{ex} = \sum_{i=1}^{2} \sum_{j>i}^{2} x_i x_j \sum_{v=0}^{n} {}^{v}L_{i,j}(x_i - x_j)^v, \tag{4}$$

where G_i^{ref} is the reference Gibbs free energy of the pure element i, i.e., Nd or Zr in the corresponding phase φ, G^{id} is the ideal mixing contribution, G^{ex} is the excess energy, x_i and x_j are the mole fractions of the constituents i, and j, respectively. ${}^{v}L_{i,j}$ is the binary interaction parameter of components i and j of an order ν. R is the gas constant, and T is the absolute temperature.

Table 2
Summary of thermodynamic parameters describing the Nd–Zr system referred to the stable element reference H^{SER} (T=298.15 K, P=100 kPa). Values are given in SI units (J, mol, and K).

Parameter	Ref.
G_{Nd}^{dhcp}, G_{Nd}^{bcc}, G_{Nd}^{liquid}, G_{Nd}^{hcp}, G_{Zr}^{bcc}, G_{Zr}^{hcp}, G_{Zr}^{liquid}	[15]
Liquid	
${}^0L_{Nd,Zr}^{liq} = +27773$	This work
${}^1L_{Nd,Zr}^{liq} = +1405$	This work
hcp	
${}^0L_{Nd,Zr}^{hcp} = +33674$	This work
dhcp	
$G_{Zr}^{dhcp} = G_{Zr}^{hcp} + 1000$	This work
${}^0L_{Nd,Zr}^{dhcp} = +38963.5$	
bcc	
${}^0L_{Nd,Zr}^{bcc} = +29664 + 7.47T$	This work

Table 3
Invariant reactions of the Nd–Zr system.

Invariant reactions	T (K)		Calc. Zr-content of phases (at% Zr)		
	Calc.	Exp.	1	2	3
L↔*bcc*-Nd+*bcc*-Zr	1271	1274 ± 10	4.0	3.0	97.0
bcc-Nd+*bcc*-Zr↔*hcp*-Zr	1167	1167 ± 10	2.2	97.8	96.6
bcc-Nd↔*hcp*-Zr+*dhcp*-Nd	1116	1118 ± 10	1.9	97.1	1.5

The thermodynamic model parameters for the binary system Nd–Zr are evaluated on the basis of the experimental information of this work using the least-squares optimization module PARROT of the multicomponent thermodynamic software Thermo-Calc [16]. First mixing parameters of *bcc* and liquid phase were assessed to fit liquidus line and eutectic reaction L↔*bcc*-Zr+*bcc*-Nd. Then mixing parameters of *hcp* and *dhcp* were evaluated to fit the other two invariant reactions *bcc*-Nd+*bcc*-Zr↔*hcp*-Zr and *bcc*-Nd↔*hcp*-Zr+*dhcp*-Nd. The final set of thermodynamic parameters was obtained by simultaneous optimisations of all mixing parameters. They are presented in Table 2. The computed Nd–Zr phase diagram using the present set of thermodynamic parameters is shown in Fig. 5 along with the experimental results. The calculated invariant equilibria are given in Table 3. The present modeling agrees well with the experimental data.

5. Conclusions

Experimental and thermodynamic assessment of the binary Nd–Zr system has been carried out. Combining gas levitation technique with *in situ* high-energy synchrotron X-ray diffraction enabled to observe directly the phase equilibria at elevated temperatures. Especially phases and their coexistence range including the liquid phase could be determined. For simple binary systems the liquidus line can be determined from the diffraction data. The experimental observations enabled to develop, a consistent thermodynamic description of the Nd–Zr phase diagram which is similar to that of the Gd–Zr system. Our results show, that the assessment of phase diagrams can get big advantages by using

in situ X-ray diffraction especially for alloy systems with high liquidus temperature, and/or reactive melts.

Acknowledgments

The authors wish to express their thanks to W. Löser and I. Kaban for valuable discussions, B. Bartusch and B. Opitz for their technical assistance. Financial support of Tohoku University Sendai for the research stay at Institute for Materials Research of one of the authors (N.M.) is gratefully acknowledged.

Appendix A. Supplementary materials

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2014.02.008.

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